

Why Use Git?

Track your work and collaborate efficiently

- Take snapshots, known as commits, of your work to track progress and easily rewind.
- Avoid file fragmentation.
- Collaborate efficiently.

```
analysis.R  
analysis_v2.R  
analysis_final.R  
analysis_final_FINAL.R  
analysis_final_FINAL_use_this_one.R
```

Git and GitHub

- Git is the software, GitHub, now run by Microsoft, is one of many services for hosting online git repositories.
- Other Git hosts:

Git Fundamentals

- Repository: A folder where Git tracks your project and its history.
- Clone: Make a copy of a remote repository on your computer.
- Stage: Tell Git which changes you want to save next.
- Commit: Save a snapshot of your staged changes.
- Branch: Work on different versions or features at the same time.
- Merge: Combine changes from different branches.
- Pull: Get the latest changes from a remote repository.
- Push: Send your changes to a remote repository.

Git Workflow

- There is a hidden directory `.git` in the main directory of a git repository.
- The `.git` directory holds all the git files, including changes that you have staged.
- Basic git workflow for a *commit* is as follows:
 - Select files to be added to a commit with `git add`.
 - Add those files to the staging area with `git commit`
 - Send those files to the cloud with `git push`.
 - Get changes made by others with `git pull`.

